Tutorial 6 Solution

Problem 1

(i) Sketch the cone generated by the columns of the matrix

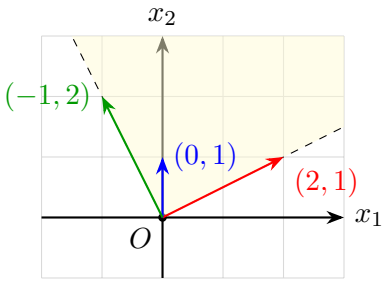
$$\boldsymbol{A} = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}.$$

(ii) What is the cone generated by just the first and third columns of the matrix $\boldsymbol{A}$?

(iii) If $\boldsymbol{b} = (1, 0)$, decide if the system $\boldsymbol{Ax} = \boldsymbol{b}$ has a solution with $\boldsymbol{x} \geq \boldsymbol{0}$.

Solution:

(i) The cone generated by the columns of the matrix looks like



(ii) The cone generated by just the first and the third columns of the matrix $\boldsymbol{A}$ is the same as the one sketched in part (a), as the second column $(0, 1)$ itself can be written as a nonnegative combination of the first and the third columns.

(iii) The Farkas alternative is the system

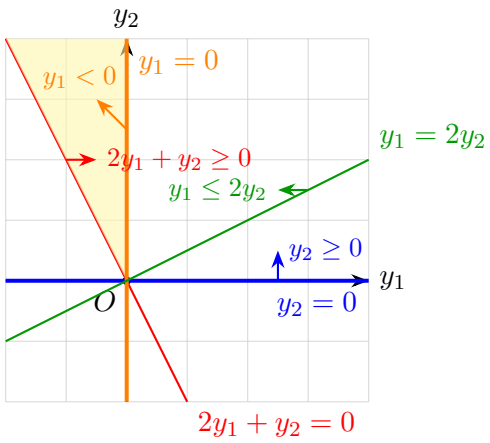
$$\begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} \geq \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = y_1 < 0.$$

As a system of inequalities the alternative is:

$$\begin{aligned} 2y_1 + y_2 &\geq 0 \\ y_2 &\geq 0 \\ -y_1 + 2y_2 &\geq 0 \\ y_1 &< 0 \end{aligned}$$

and the feasible region of the system is graphed in the picture below in yellow.

We see that the feasible region is non-empty, implying that the alternative has a solution. By the Farkas lemma, the system $\boldsymbol{Ax} = \boldsymbol{b}$ *does not* have a non-negative solution.



Problem 2
Use the Farkas lemma to decide if the following system has a non-negative solution:

$$\begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Solution: The Farkas alternative is

$$\begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} \geq \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, \quad \text{and} \quad y_1 + y_2 < 0.$$

As a system of inequalities the alternative is:

$$\begin{aligned} 4y_1 + y_2 &\geq 0, \\ y_1 + 0y_2 &\geq 0, \\ -5y_1 + 2y_2 &\geq 0, \\ y_1 + y_2 &< 0. \end{aligned}$$

There can be no solution to this system. The second inequality requires that $y_1 \geq 0$. Combining this with the last inequality we conclude that $y_2 < 0$. But $y_1 \geq 0$ and $y_2 < 0$ contradict $-5y_1 + 2y_2 \geq 0$. Thus, the original system has a non-negative solution.

Problem 3

Let $\boldsymbol{A}$ be a $m \times n$ matrix, and let $\boldsymbol{b} \in \mathbb{R}^m$. Prove, using the Farkas Lemma, the following:

(1) Either the system $\boldsymbol{Ax} \geq \boldsymbol{b}$ has a *non-negative* solution, or there is a *non-negative* $\boldsymbol{y} \in \mathbb{R}^m$ such that $\boldsymbol{yA} \leq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} > 0$, but not both.

(2) (**Extra Practice Problem**) Either the system $\boldsymbol{Ax} \leq \boldsymbol{b}$ has a solution, or there is a *non-negative* $\boldsymbol{y} \in \mathbb{R}^m$ such that $\boldsymbol{yA} = \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} < 0$, but not both.

(*Hint.* Recall how we found the Farkas alternative for the system $\boldsymbol{Ax} \leq \boldsymbol{b}$ and $\boldsymbol{x} \geq \boldsymbol{0}$ in the lecture.)

Solution:

(1) There are (at least) two ways to solve this part. One way is the standard way: we subtract a surplus variable $s_1 \geq 0$ to each of the inequalities, $\sum_{j=1}^{n} a_{ij}x_j \leq b_i$ to convert it to an equality $\sum_{j=1}^{n} a_{1j}x_j - s_i = b_i$. For notational convenience, let $\boldsymbol{s} \equiv (s_1, s_2, \ldots, s_m)$ be the *surplus vector*; we have $\boldsymbol{s} \geq \boldsymbol{0}$. Determining the solvability of $\boldsymbol{Ax} \geq \boldsymbol{b}$ for $\boldsymbol{x} \geq \boldsymbol{0}$ is equivalent to solvability of $\boldsymbol{Ax} - \boldsymbol{s} = \boldsymbol{b}$ where $\boldsymbol{x} \geq \boldsymbol{0}$ and $\boldsymbol{s} \geq \boldsymbol{0}$, which is in turn equivalent to solvability of $\boldsymbol{Ax} - \boldsymbol{Is} = \boldsymbol{b}$, where $\boldsymbol{I}$ denotes the $m \times m$ identity matrix.

Set $\boldsymbol{B} \equiv [\boldsymbol{A}| - \boldsymbol{I}]$ and $\boldsymbol{z} \equiv (\boldsymbol{x}, \boldsymbol{s})$, the system can be written as $\boldsymbol{Bz} = \boldsymbol{b}$ and $\boldsymbol{z} \geq \boldsymbol{0}$. Applying the Farkas lemma to this system, we get $\boldsymbol{y'B} \geq 0$ and $\boldsymbol{y'} \cdot \boldsymbol{b} < 0$. Since $\boldsymbol{B} = [\boldsymbol{A}| - \boldsymbol{I}]$, this is equivalent to

$$\boldsymbol{y'A} \geq \boldsymbol{0}, \quad \boldsymbol{y'} \leq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y'} \cdot \boldsymbol{b} < 0.$$

Finally, since we need to prove a statement about a non-negative $\boldsymbol{y} \in \mathbb{R}^m$, we set $\boldsymbol{y} \equiv -\boldsymbol{y'} \geq 0$, and the Farkas alternative is

$$\boldsymbol{yA} \leq \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} > 0.$$

The other approach, which is arguably much more straightforward but didactically less helpful, is essentially a trick: note that $\boldsymbol{Ax} \geq \boldsymbol{b}$ can be written as $-\boldsymbol{Ax} \leq -\boldsymbol{b}$; define $\boldsymbol{C} \equiv -\boldsymbol{A}$ and $\boldsymbol{d} \equiv -\boldsymbol{b}$, the system can be written as $\boldsymbol{Cx} \leq \boldsymbol{d}$ and $\boldsymbol{x} \geq 0$. Then as we discussed in the lecture, the Farkas alternative for the system is $\boldsymbol{yC} \geq \boldsymbol{0}$, $\boldsymbol{y} \geq \boldsymbol{0}$, and $\boldsymbol{y} \cdot \boldsymbol{d} < 0$. By the definition of $\boldsymbol{C}$ and $\boldsymbol{d}$, this is equivalent to

$$\boldsymbol{yA} \leq \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} > 0.$$

(2) Recall that in the lecture we mentioned that for a variable, x_j, that is unrestricted in sign, we can replace it by two non-negative variables z_j and z'_j by setting $x_j = z_j - z'_j$. Since $\boldsymbol{x}$ is unrestricted in sign, applying this to every component of $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$, we get

$$\boldsymbol{x} = \boldsymbol{z} - \boldsymbol{z'},$$

where $\boldsymbol{z} \equiv (z_1, z_2, \ldots, z_n) \geq \boldsymbol{0}$, and $\boldsymbol{z'} \equiv (z'_1, z'_2, \ldots, z'_n) \geq \boldsymbol{0}$. Now the system $\boldsymbol{Ax} \leq \boldsymbol{b}$ is equivalent to the system

$$\boldsymbol{A}(\boldsymbol{z} - \boldsymbol{z'}) \leq \boldsymbol{b}, \quad \boldsymbol{z} \geq 0, \quad \text{and} \quad \boldsymbol{z'} \geq 0;$$

and $\boldsymbol{A}(\boldsymbol{z} - \boldsymbol{z'}) \leq \boldsymbol{b}$ can also be written as $\boldsymbol{Az} - \boldsymbol{Az'} \leq \boldsymbol{b}$.

Set $\boldsymbol{D} \equiv [\boldsymbol{A}| - \boldsymbol{A}]$ and $\boldsymbol{w} \equiv (\boldsymbol{z}, \boldsymbol{z'})$, the system can be expressed as

$$\boldsymbol{Dw} \leq \boldsymbol{b} \quad \text{and} \quad \boldsymbol{w} \geq \boldsymbol{0}.$$

The Farkas alternative of this system is

$$\boldsymbol{yD} \geq \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} < 0.$$

Since $\boldsymbol{D} = [\boldsymbol{A}| - \boldsymbol{A}]$, this system is equivalent to

$$\boldsymbol{yA} \geq \boldsymbol{0}, \quad \boldsymbol{yA} \leq \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} < 0,$$

which can be further simplified to

$$\boldsymbol{yA} = \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} < 0.$$

Problem 4

Consider the system of equations

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &\leq 5, \\ x_1 + 3x_2 - 2x_3 &\geq 7, \\ x_1 + x_2 + x_3 &\leq 2, \\ x_1 - 2x_2 - 3x_3 &\geq 3, \\ x_1 &\geq 0, \\ x_2 &\geq 0, \\ x_3 &\geq 0. \end{aligned}$$

Write down its Farkas alternative.
(*Hint.* Try to write down this system in matrix form first. How are you going to deal with the first four inequalities in doing that?)

Solution: The second and the fourth inequality can be written as

$$-x_1 - 3x_2 + 2x_3 \leq -7 \quad \text{and} \quad -x_1 + 2x_2 + 3x_3 \leq -3,$$

respectively. Therefore, the system can be written as

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \begin{pmatrix} 5 \\ -7 \\ 2 \\ -3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The system can be written as $\boldsymbol{Ax} \leq \boldsymbol{b}$ and $\boldsymbol{x} \geq \boldsymbol{0}$; according to our discussions on inequality systems in the lecture, the Farkas alternative takes the form of $\boldsymbol{yA} \geq \boldsymbol{0}$, $\boldsymbol{y} \geq \boldsymbol{0}$, and $\boldsymbol{y} \cdot \boldsymbol{b} < 0$. Thus, the Farkas alternative is

$$\begin{pmatrix} y_1 & y_2 & y_3 & y_4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \geq \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} y_1 & y_2 & y_3 & y_4 \end{pmatrix} \geq \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix},$$

and $5y_1 - 7y_2 + 2y_3 - 3y_4 < 0$.